Catering & Event Ordering Platform

Phase 1 Database Design

Version: 1.0

Status: Approved for MVP Development

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1. Purpose

This document defines the Phase 1 database architecture for the Catering & Event Ordering Platform.

The design supports:

- Customer event planning
- Package-based catering selection
- Menu customization within package constraints
- Dynamic package pricing
- Online payments
- Order management
- Administrative operations

The schema is designed to support future enhancements including:

- Custom package creation
- Mobile applications
- Loyalty programs
- Vendor management
- Marketplace functionality

without requiring major database redesign.

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2. Business Model Overview

The platform follows a Package-Driven Catering Model.

Example:

Silver Package

₹499 Per Plate

Includes:

- 2 Starters
- 3 Main Courses
- 1 Dessert

Customers can:

- Select dishes from available options
- Replace dishes within a category
- Upgrade to premium dishes
- Pay additional charges where applicable

The final order price is calculated as:

Final Package Price = Base Package Price + Customization Charges

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3. Database Architecture

Core Domains

Customer Management

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Event Planning

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Package Management

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Menu Management

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Order Management

↓

Payments

↓

Administration

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4. Entity Relationship Overview

Users

├── Addresses

├── Events

│ └── Orders

│ └── Selected Menu Items

│ └── Payments

Packages

├── Package Versions

├── Category Rules

├── Allowed Menu Items

└── Pricing Rules

Menu

├── Categories

└── Menu Items

Administration

├── Admin Users

└── Order Status History

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5. Customer Domain

Users

Stores customer accounts.

Primary Responsibilities:

- Login

- Customer Profile
- Event Ownership
- Order Ownership

Primary Key

- id

Indexes

- mobile_number (Unique)

User Addresses

Stores customer delivery and event locations.

Supports:

- Multiple addresses
- Default address selection

Foreign Keys

- user_id → users.id

Indexes

- user_id

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6. Menu Domain

Menu Categories

Defines food groupings.

Examples:

- Starter
- Main Course
- Dessert
- Beverage

Primary Key

- id

Indexes

- name (Unique)

Menu Items

Master catalog of dishes.

Examples:

- Veg Manchuria
- Paneer Tikka
- Veg Biryani
- Gulab Jamun

Key Attributes:

- Base Price
- Category
- Veg/Non-Veg
- Active Status

Foreign Keys

- category_id → menu_categories.id

Indexes

- category_id
- is_active

Unique Constraint

- category_id + name

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7. Package Domain

Packages

Commercial offerings available to customers.

Examples:

- Silver Package
- Gold Package
- Premium Package

Key Attributes:

- Package Name
- Base Price Per Plate
- Minimum Guest Count

Indexes

- name (Unique)
- is_active

Package Versions

Supports package evolution over time.

Example:

Silver Package

Version 1 → ₹499

Version 2 → ₹549

Existing orders continue referencing the original version.

Foreign Keys

- package_id → packages.id

Unique Constraint

- package_id + version_no

Package Category Rules

Defines category selection limits.

Examples:

Starter → Select 2

Main Course → Select 3

Dessert → Select 1

Foreign Keys

- package_version_id
- category_id

Unique Constraint

- package_version_id + category_id

Package Menu Items

Defines dishes available within a package.

Example:

Silver Package

Starter Options:

- Veg Manchuria
- Paneer Tikka
- Spring Roll

Foreign Keys

- package_version_id
- category_id
- menu_item_id

Unique Constraint

- package_version_id + menu_item_id

Package Menu Item Pricing

Defines package-specific pricing rules.

Example:

Menu Item Price = ₹80

Package Included Value = ₹60

Customer Upgrade = ₹20

Customization Charge is calculated dynamically.

Foreign Keys

- package_version_id
- menu_item_id

Unique Constraint

- package_version_id + menu_item_id

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8. Event Domain

Events

Represents a planned catering event.

Examples:

- Birthday Party
- Wedding
- Corporate Event
- Housewarming

Stores:

- Event Date
- Guest Count
- Package Selection
- Event Address

Foreign Keys

- user_id
- package_version_id
- address_id

Indexes

- user_id
- event_date
- status

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9. Order Domain

Orders

Commercial transaction generated from an event.

Stores pricing snapshots including:

- Package Price
- Customization Charges
- Final Per Plate Price
- Total Amount

Foreign Keys

- user_id
- event_id

Indexes

- order_number (Unique)
- user_id
- order_status
- payment_status
- created_at

Order Selected Items

Stores the exact menu selected for the order.

Important:

Order pricing must never be recalculated from master tables.

A pricing snapshot is stored.

Stored Attributes:

- Menu Item Name
- Menu Price
- Included Price
- Adjustment Amount

Foreign Keys

- order_id
- category_id
- menu_item_id

Indexes

- order_id

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10. Payment Domain

Payments

Stores all payment transactions.

Supports:

- Razorpay
- Refunds
- Future payment gateways

Foreign Keys

- order_id

Indexes

- order_id
- payment_status
- gateway_payment_id

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11. Administration Domain

Admin Users

Administrative access for platform operations.

Roles:

- ADMIN
- OPERATIONS

Indexes

- email (Unique)

Order Status History

Tracks every status change.

Example:

Pending

↓

Confirmed

↓

In Progress

↓

Delivered

Supports:

- Audit Trail
- Customer Support
- Reporting

Foreign Keys

- order_id
- changed_by

Indexes

- order_id
- created_at

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12. Recommended Status Definitions

Order Status

DRAFT

PENDING_PAYMENT

CONFIRMED

IN_PROGRESS

READY_FOR_DELIVERY

DELIVERED

CANCELLED

Payment Status

PENDING

PAID

FAILED

REFUNDED

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13. Performance & Indexing Strategy

High Traffic Queries

Customer Orders

Search by:

- user_id
- created_at

Upcoming Events

Search by:

- event_date
- status

Admin Dashboard

Search by:

- order_status
- payment_status
- created_at

Payment Reconciliation

Search by:

- gateway_payment_id
- payment_status

Package Configuration

Search by:

- package_version_id
- category_id

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14. Future Expansion Readiness

The design supports future enhancements without structural redesign.

Phase 2

- Saved Menus
- Repeat Orders
- Notifications
- Advanced Reporting

Phase 3

- Custom Package Builder
- Loyalty Programs
- Referral Programs

Phase 4

- Vendor Management
- Multi-Vendor Marketplace
- Mobile Applications
- AI-Based Menu Recommendations

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15. Final Design Principles

The database has been designed to:

- Maintain historical pricing accuracy
- Support package versioning
- Support package customization
- Scale to future custom-package creation
- Support reporting and analytics
- Minimize future schema migrations
- Support responsive web and future mobile applications

This schema serves as the authoritative Phase 1 data model for the Catering & Event Ordering Platform MVP.